DO YOU KNOW?

A preservative is a substance that is added to food to inhibit and retard the growth of microorganisms and helps in the process of preservation.

Principles of Food Preservation

1. ***Prevention or delay of microbial decomposition.***
   a. By keeping out microorganisms and preventing contamination from pathogens. It involves applying the strictest rules to minimize the risk of infection *(asepsis)*.
   b. Removal of microorganism through usage of membrane which retains microorganisms *(filtration)*.
   c. By hindering the growth and activity of microorganisms *(refrigeration, dehydration, addition of chemical preservatives)*.
   d. By killing microorganisms *(boiling, irradiation)*.
2. ***Prevention or delay of self decomposition of food***
   a. By destruction or inactivation of enzymes *(blanching)*.
   b. By prevention or delay of chemical reactions *(anti oxidants)*.
3. ***Prevention of damage caused by mechanical causes, insects and rodents.***

4.2 PRESERVATION METHODS

Many foods cannot be stored as such and need to undergo a treatment or a technique which helps to prevent spoilage. The techniques adopted to preserve the foods are grouped into ***traditional techniques and modern industrial techniques.*** *Traditional techniques* include curing, freezing, boiling, heating, sugaring, pickling, canning, smoking, salting and fermentation. *Modern industrial technique* involve pasteurization, vacuum packing, artificial food additives and irradiation.

In this chapter, preservation techniques are discussed under the following headings. They are use of low temperature, use of high temperature, dehydration, use of chemical preservatives and preservation by high osmotic pressure.

Activity 1

Boil sliced potatoes in water for three minutes. Observe the changes that take place in the texture, colour and appearance.

4.2.1 Preservation of Foods with Low Temperature

Use of low temperature reduces the microbial activity and enzyme activity thus prolongs shelf life of foods. Two different temperatures are employed in low temperature namely *chilling temperature and freezer temperature.*

4.2.1.1 Chill Storage

Chill temperature is just above the refrigerated temperature. In chilling fish, the temperature is reduced to freezing point of water. Chill temperature delays both bio-chemical and bacteriological changes. The deteriorative changes are retarded when low temperature is maintained. Hence the shelf life of food is improved and this ensures preserving natural and functional properties of food. Storage at *−1°C and −4 °C* can provide stability in the presence of food preservation.

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